Assignment

1. Find $f(t)$ .given Fourier cosine transform is $\mathcal{F}_C(\omega) = \begin{cases} 1 & for\ 0 < \omega < k \\ 0 & for\ \omega > k \end{cases}$.

2. Find the function which yields $\mathcal{F}_C(\omega) = \frac{sink\omega}{\omega}$ as its Fourier cosine transform.

$$f(t) = \begin{cases} 1 & for\ \ t < |k| \\ 0 & for\ t > |k| \end{cases}$$